

Statistics 801
Final Examination (100 points)

Name _____

1. The price of a laptop of a particular make and configuration among dealers nationwide is assumed to have a Normal distribution with mean $\mu = \$1200$ with standard deviation $\sigma = \$50$.
 - (a) (6) What is the probability that a laptop of the same make, chosen randomly from a dealer, will cost between \$1136 and \$1308 ? (Show your work).

 - (b) (6) What is the probability that the average price of 25 laptops of the same make, chosen randomly from a dealer, will be greater than \$1225? (Show your work).

2. To estimate the mean potency μ of an antibiotic in a production batch, a manufacturer takes potency readings of 14 random samples, giving $\bar{y} = 8.67$, and $s^2 = 0.56$.
 - (a) (6) Estimate the mean potency for the batch using a 95% confidence interval.

- (b) (6) According to the industry standard, the standard deviation of the potency for this drug, σ , must not exceed 0.7. *State and test* the hypotheses that this batch does not meet the industry standard. Use $\alpha = .05$. Don't forget your conclusion and decision.

3. Brands of a type of electrical wire produced by two different manufacturers are being studied to compare their resistivity characteristics. The values of resistivity measurements obtained by the experimenter suggests that they can be considered to be samples from normal distributions and are used to calculate the following statistics:

Brand A			Brand B		
$n_1 = 8$	$\bar{y}_{1.} = 0.140$	$s_1^2 = 0.000007$	$n_2 = 9$	$\bar{y}_{2.} = 0.145$	$s_2^2 = 0.00002$

Assuming that the population means are μ_1 and μ_2 and the population variances are σ_1^2 and σ_2^2 , respectively, answer the following questions:

- (a) (6) Test the hypothesis $H_0 : \sigma_1^2 = \sigma_2^2$ vs. $H_a : \sigma_1^2 \neq \sigma_2^2$. Use $\alpha = .05$ and show work.

- (b) (4) Now assume $\sigma_1^2 = \sigma_2^2$. Compute the standard error of the difference in the sample means $\bar{y}_{1.} - \bar{y}_{2.}$.

- (c) (6) Use the standard error from part (b) to construct a statistic to test the research hypothesis that Brand B wires have larger mean resistivity. State the null and alternative, the rejection region, and your decision using $\alpha = .05$. Show work.

- (d) (6) Use the standard error from part (b) to compute a 90% confidence interval for $\mu_1 - \mu_2$. Test the hypothesis in part (c) using this interval, stating the α -level of the test.

4. An experiment was conducted to determine the relationship between the amount of warping in thin strips, (y), made from a particular alloy and temperature, (x) . A simple linear regression model of the form $y = \beta_0 + \beta_1 x + \epsilon$ was first fitted to the data.

Temperature ($^{\circ}C$) (x)	Amount of Warping (y)	$\bar{y}_i$	$y_i - \bar{y}_i$	$\sum(y_i - \bar{y}_i)^2$	df
15	11, 11, 14	12			
20	14, 12, 13	13			
25	14, 12, 16	14			
30	18, 14, 16	16			
35	21, 19, 17	19			
40	24, 20, 22	22			
45	24, 23, 28	25			
50	31, 30, 32	31			

The following statistics have also been computed from the above data:

$$S_{xx} = 3150.0 \quad S_{yy} = 980.0 \quad S_{xy} = 1650.0$$

- (a) (10) Complete the analysis of variance table for the regression below. Use the **F-ratio** from the anova table to test $H_0 : \beta_1 = 0$ vs. $H_a : \beta_1 \neq 0$, at $\alpha = .05$. Show work and state your conclusion.

<u>Source</u>	<u>d.f.</u>	<u>SS</u>	<u>MS</u>	<u>F</u>
Regression				
Error				
Total	23			

- (b) (6) Compute the t-statistic to test the hypothesis that $H_0 : \beta_1 = 0$ vs. $H_a : \beta_1 \neq 0$. Test this hypothesis using $\alpha = .05$. State the rejection region and your decision clearly. Show work.

- (c) (10) Perform the lack of fit test by completing the following table and state your conclusion. Use $\alpha = .05$.

<u>Source</u>	<u>d.f.</u>	<u>SS</u>	<u>MS</u>	<u>F</u>
Lack of Fit				
Pure Error				
Total Error				

- (d) (3) Calculate the coefficient of determination.

(e) (3) Interpret the coefficient of determination in terms of the problem.

5. An agronomist wanted to see whether four chemical treatments were different in the number of soybean seeds that fail to emerge out of 100 seeds planted. She uniformly planted 100 soybean seeds treated with one of the four treatments in a randomly assigned plot in 5 different fields. She then counted the number of seeds that failed to emerge out of 100 seeds.

(a) (3) Write out the linear additive model for this study and define the terms. State the assumptions.

(b) (2) What type of experimental design is it? How do you know? Justify your answer.

(c) (7) Part of the ANOVA table is shown below. Fill in the missing values in the ANOVA table.

Source	degree of freedom	Sum of Squares	Mean of Squares	F-value	p-value
Field			21.925		0.0001
Chemical Treatment			21.7833		0.0001
Error	12		2.325	-	-
Total			-	-	-

(d) (2) Based on the p-value and $\alpha = 0.05$, what can you conclude about the original research question?

(e) (2) Does it appear that blocking by field was effective? Support your answer. $\alpha = 0.05$.

6. What type of analyses (Contingency Table/Chi-square test or ANOVA or Regression) would you use for the following set of variables. Give an example for each.

(a) (2) Quantitative dependent, categorical independent

(b) (2) Categorical dependent, categorical independent

(c) (2) Quantitative dependent, quantitative independent

7. **BONUS QUESTIONS.** Each question is 1 point.

- (a) **(T/F)** In the analysis of covariance we compare the sum of squares of a full model with the sum of squares of a reduced model.
- (b) **(T/F)** The analysis of a factorial design is a better way to look at the interaction among treatments than the completely randomized design.
- (c) **(T/F)** The advantage of random forest over the linear regression is that the involving model is much simpler.
- (d) **(T/F)** The permutation test is different than the randomization test because in the permutation test we sample without replacement, and in the randomization test we sample with replacement.
- (e) **(T/F)** The Mann-Whitney test can only be used when the data are continuous.